

Skju - Seismic Stations Network

todo

Analysis of time-series data boils down to understanding how events that happened in the past will affect future events. [1]

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1. Skju Architecture

This section discusses the high-level design for Skju's architecture.

2. Simulation

In seismology, attenuation describes how seismic wave amplitude decreases with distance from the earthquake source.

Skju model them thought combination of two physical effects:

- Geometric spreading — surface waves, spreading in 2D along the surface

$$A \sim \frac{1}{\sqrt{r}}$$

- Anelastic attenuation — surface media absorbs waves energy.

Often (?) modelled as exponential decay on top of geometric term.

$$A(r) = A_0 \cdot \frac{1}{r} \cdot \exp\left(-\pi \cdot f \cdot \frac{r}{Q \cdot v}\right) \quad (1)$$

A_0 — source amplitude (depends on magnitude) r — distance from source f — frequency of the wave Q — quality factor of the medium v — wave velocity

A_0 is tied to earthquake magnitude.

The Richter scale relationship is logarithmic:

$$A_\emptyset = 10^{M \cdot c}$$

3. Sensors and Seismic Stations Identification

Reliable node identification is a foundational requirement in distributed sensor networks, underpinning message routing, data provenance, fault isolation, and deduplication of telemetry. In safety-critical contexts such as seismic monitoring, identifier integrity directly affects the reliability of event attribution.

The Skju system comprises two classes of hardware nodes:

- field sensors built on the Nordic Semiconductor nRF52 series
- and seismic stations built on the nRF91 series.

To uniquely identify each node across the network, we evaluated

- auto-increment integer counters,
- UUIDv4,
- and IMEI printed by Nordic on the nrf9160dk product's label

all of which would have required identifiers to be programmed into hardware during production.

We ultimately adopted the hardware-native DEVICEID register available on Nordic Semiconductor chips.

Each DEVICEID value is generated randomly during chip production using a FIPS-approved random number generator, yielding a per-pair collision probability of (5.42×10^{-20}) effectively negligible at the scale of any foreseeable deployment. Therefore DEVICEID can be reliably used as MQTT client ID.

4. Skatch

1. Depending on the region of Earth you are living in

earth quake can be serious problem causing (TODO: provide a reference) money in damages of infrastructure health expenses and expenses non recoverable - human death and casualties.

According to ...

5. Electrical Characteristis

Current consumption @ 3.7 V:

- Power saving mode (PSM) floor current: 2.7 μA
- eDRX @ 81.92s: 18 μA in Cat-M1, 37 μA in Cat-NB1 (UICC included)

[2]

6. Naive Earth Quake Detection

multisite systems at low cost with low latency

We successfully developed a low-latency, Bluetooth low energy (BLE)-based data alignment method, implemented in the BLE application layer, making it transferable between manufacturer devices. The time synchronization method was tested on two commercial BLE platforms by inputting common sinusoidal input signals (over a range of frequencies) to evaluate time alignment performance between two independent peripheral nodes.

7. Time Synchronization in Sensors's Network

1. Describe the motivation for time synchronization.
2. Describe the prior art [3].
3. Describe the approach we took.

8. Heim Service

Heim Service is ment to be an interal service and a source of truth for Seismic Stations' metadata.

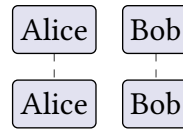


Figure 1: A curious figure.

Yet, there is one exception — Simulator iPadOS app. The app is allow to read and write to Heim from outside of protected perimeter.

</> Protol Buffer Definition

```
// some protocol buffers definition
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Bibliography

- [1] A. Nielsen, “Practical Time Series Analysis: Prediction with Statistics and Machine Learning,” 2020.
- [2] Nordic Semiconductor, “nRF9160 Product Specification v2.2.” 2024.
- [3] Y. Guo, Z. Wang, G. Zhang, and B. Li, “An Efficient and Robust Real-Time Video De-Interlacing Method for Low-Power Devices,” *Sensors*, vol. 23, no. 8, 2023, doi: 10.3390/s23083954[°].

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